

Group Project (2nd Theme)

Learning outcome:

The main objective of this group project is to provide practice to the students in gathering information, defining problems, critical thinking, creativity, teamwork collaboration, and communication. The project addresses the following learning Collaborative Problem Solving (CPS) outcomes:

- CPS3-2: Analyze algorithms and techniques to design and optimize computing solutions.
- CPS3-5: Demonstrate effective interaction with various stakeholders
- CPS3-10: Apply an entrepreneurial mindset in providing solutions

Project Description

The full mark of the project is 20% of the total assessment of this course. Form a team with 6 or 7 members per group.

1. **Identify a Global Scenario:** Identify a specific challenge in raising awareness of sexual harassment or a gap in developing anti-sexual harassment initiatives that resonates on a campus or global scale. You must develop an AI application using software-based tools to impact this scenario.
 - **Theme: Intelligent Anti-Harassment Solutions** (Software-only; focus on mobile-first or web-based tools that track safety resources or simplify reporting pathways without requiring specialized hardware).
 - **Requirements:** Explain the AI algorithms used (e.g., NLP for sentiment analysis of policy or Decision Trees for reporting roadmaps), how they solve the problem, and provide a critical review of the algorithm's suitability for supporting survivors in a university setting.
 - **Report:** The report must contain links to the developed video, a technical explanation of the AI logic, and a full list of references.
 - **Develop a Video:** Produce a high-impact video to demonstrate your solution, its ability to increase campus safety awareness, and its potential to attract humanitarian partners.

Two documents must be delivered:

1. **Manual in developing the application:** Technical documentation (system architecture, environment setup, and code logic).
2. **Manual how to use the application:** A user-centric guide designed for end-user base.

Example of scenarios:

- Peer-to-Peer Verbal Harassment
- Digital Stalking and Cyber-Harassment
- Physical Boundary Violations in Crowded Areas
- Active Bystander Intervention in Student Clubs

- Navigation of University Redressal Mechanisms
- Identifying Grooming Behavior
- Safety Resource Accessibility
- Intelligent Sexual Harassment Self-Classification Tool

Projects timeline

Submission	Description
Week 8 (17/5/2026)	<u>Project Proposal</u> 4-5 pages proposal contains the following information: • project team members • project title and objective • problem statement from 5 years latest articles • contribution of the project • precise description of the question you are trying to answer and how you will answer it (what data you will collect or from where you extract it) • methodology • reading list (any papers/sources you read) Record a video for your proposal slide presentation
Week 12 and Week 13	<u>Two minutes project descriptions</u> Describing your project to the class using slide presentation and demonstrate the project. Include a brief description of the application you developed, the problem solution, its impact on the community, etc. Make it exciting and creative!
Week 13	<u>Post your application at media social</u> Present your AI application to Youtube and provide the link in your documents and presentation slides
	<u>Document submission</u> Two document manuals of the application